

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Pharmacy Management System

Requirements Baseline – Version 1.0

DOCUMENT INFORMATION

Project	Pharmacy Management System (PMS)
Project Reference	JF/ICT/24/30
Document	Software Requirements Specification
Version	1.0
Baseline	Requirements Baseline v1.0
Development Approach	Incremental Development with Controlled Change Management
Technology – Frontend	HTML, CSS, JavaScript
Technology – Backend	Node.js, Express.js
Technology – Database	MySQL
Development Environment	Visual Studio Code, XAMPP / MySQL Server
Status	Proposed for Review
Prepared By	Student Project Team
Date	_____

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the Pharmacy Management System. The document establishes the agreed requirements that will be used as the reference for system analysis, system design, database design, software development, testing, release activities, and Software Change Management.

The requirements contained in this document represent the initial proposed scope. Once reviewed and approved by the relevant stakeholders, this document will become the Requirements Baseline v1.0. Any new requirement introduced after the Version 1.0 baseline has been approved shall be handled through the Software Change Management process using a formal Change Request (CR).

1.2 Project Background

Pharmacies manage a large amount of information including medicines, stock quantities, batch details, expiry dates, suppliers, sales transactions, and user information. When these activities are handled manually or through separate files, staff may spend considerable time searching for information, updating stock, checking expiry dates, and preparing sales records.

The proposed Pharmacy Management System will provide a centralized web-based platform for managing essential pharmacy operations. Version 1.0 focuses on authentication, medicine and inventory management, sales management, supplier information, and role-based access for Admin, Pharmacist, and Pharmacy Staff.

2. PROBLEM STATEMENT

- Pharmacy information may be stored in separate files or manual records.
- Searching for medicine and stock information can be time-consuming.
- Manual stock updates can result in quantity errors.
- Expiry dates may be difficult to monitor consistently.
- Sales records can be difficult to maintain and retrieve.
- Duplicate or inconsistent records may be created.
- Access to pharmacy information may be difficult to control.
- Manual preparation of stock and sales information requires additional effort.

The proposed system aims to address these problems by providing a centralized web-based platform for managing pharmacy records and essential daily operations.

3. PROJECT OBJECTIVES

- To develop a web-based system for managing pharmacy information.
- To provide secure user authentication and role-based access.
- To allow authorized users to add, view, update, and manage medicine records.
- To maintain medicine stock quantities and batch information.
- To record pharmacy sales transactions accurately.
- To maintain supplier information.
- To support monitoring of low-stock and expired/near-expiry medicines.
- To reduce manual pharmacy record management.
- To improve the accuracy and consistency of pharmacy information.
- To provide a responsive and user-friendly interface.
- To demonstrate controlled software evolution using Software Change Management practices.

4. PROJECT SCOPE

4.1 In Scope for Version 1.0

The Version 1.0 system shall provide:

Authentication & Authorization

- User login
- User logout
- Credential validation
- Basic authorization
- Role-based access for Admin, Pharmacist, and Pharmacy Staff

Medicine Management

- Register new medicines
- View medicine list
- Update medicine information
- Delete medicine records
- Search medicines
- Medicine category management

Supplier Management

- Add supplier information
- View supplier details
- Update supplier information
- Delete supplier records
- Manage supplier details

Inventory Management

- Medicine batch management
- Medicine expiry-date management
- Stock quantity management
- Automatic stock deduction after completed sales
- Low-stock monitoring
- Track medicine availability

Sales Management

- Record sales transactions
- Add medicines to sales
- Manage sales item and quantity
- Calculate sales transaction information
- Complete sales transactions
- View sales records

Dashboard & Validation

- Basic dashboard
- Display summary information
- Display stock information
- Display low-stock information
- Basic input validation
- Error messages for invalid inputs

Technical Scope

- HTML, CSS, and JavaScript frontend
- PHP backend
- MySQL database
- Responsive web interface
- Database-based data storage
- Role-based access control
- Basic system security and validation

4.2 Out of Scope for Version 1.0

- Online medicine ordering
- Online payment gateway integration
- Mobile application
- Advanced accounting and payroll
- Insurance claim processing
- Advanced analytics and business intelligence
- External laboratory integration
- Automatic supplier purchasing
- SMS/email notification integration
- Multi-branch enterprise deployment
- Advanced audit-log management
- AI-based diagnosis or prescription generation

These features may be introduced in future releases through approved Change Requests.

5. STAKEHOLDERS

Stakeholder	Role	Interest / Responsibility
Administrator	System User	Manage users, medicines, suppliers, stock, and system information
Pharmacist	Business User	Manage medicines, stock, sales, and pharmacy operations
Pharmacy Staff	Operational User	Perform permitted daily pharmacy and sales activities
Pharmacy Management	Management Stakeholder	Monitor pharmacy operations and system outcomes
IT Administrator	Technical Stakeholder	Maintain and support the system
Project Team	Development Team	Analyze, design, develop, test, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Monitor project progress and SCM practices

6. USER ROLES

6.1 Administrator

- Log in and log out.
- Manage system users and roles.
- Add, view, update, and manage medicine records.
- Manage suppliers and categories.
- View and manage stock information.
- View sales records and basic reports/dashboard information.

6.2 Pharmacist

- Log in and log out.
- View and manage permitted medicine information.
- Manage stock and batch information.
- Record and manage sales transactions.
- Search medicines.
- Monitor low-stock and expiry information.

6.3 Pharmacy Staff

- Log in and log out.
- Search and view permitted medicine information.
- Record sales transactions according to assigned permissions.
- View relevant stock information.
- Cannot access restricted administrative functions.

7. FUNCTIONAL REQUIREMENTS

7.1 Authentication Requirements

FR-01 – User Login

The system shall allow an authorized user to log in using a valid username and password.

FR-02 – Login Validation

The system shall validate the credentials provided by the user.

FR-03 – Invalid Login Handling

The system shall display an appropriate error message when invalid login credentials are provided.

FR-04 – User Logout

The system shall allow an authenticated user to log out of the system.

FR-05 – Role Authorization

The system shall provide access to functions according to the user's assigned role.

7.2 Medicine Management Requirements

FR-06 – Add Medicine

The system shall allow authorized users to add a new medicine record.

FR-07 – Medicine Identifier

The system shall maintain a unique identifier for each medicine record.

FR-08 – View Medicine List

The system shall allow authorized users to view registered medicines.

FR-09 – View Medicine Details

The system shall allow authorized users to view details of a selected medicine.

FR-10 – Update Medicine

The system shall allow authorized users to update medicine information.

FR-11 – Delete Medicine

The system shall allow authorized users to delete or deactivate a medicine record according to permissions.

FR-12 – Search Medicine

The system shall allow permitted users to search for medicines using relevant information such as name, code, or category.

7.3 Inventory Requirements

FR-13 – Stock Management

The system shall maintain stock quantities for medicines.

FR-14 – Batch Information

The system shall allow batch-related information to be recorded for medicines where required.

FR-15 – Expiry Date

The system shall store and display medicine expiry dates.

FR-16 – Low Stock Monitoring

The system shall identify medicines whose stock quantity is at or below the defined minimum level.

FR-17 – Expiry Monitoring

The system shall allow authorized users to identify expired or near-expiry medicines.

FR-18 – Stock Update

The system shall update stock quantities when permitted stock operations are completed.

7.4 Supplier Requirements

FR-19 – Add Supplier

The system shall allow authorized users to add supplier information.

FR-20 – View Supplier

The system shall allow authorized users to view supplier information.

FR-21 – Update Supplier

The system shall allow authorized users to update supplier information.

FR-22 – Search Supplier

The system shall allow authorized users to search supplier records.

7.5 Sales Requirements

FR-23 – Create Sale

The system shall allow permitted users to record a medicine sale.

FR-24 – Sales Items

The system shall record the medicines and quantities included in a sale.

FR-25 – Stock Deduction

The system shall reduce available stock quantities when a sale is successfully completed.

FR-26 – Sale Total

The system shall calculate the total amount of a sales transaction based on the recorded items and prices.

FR-27 – View Sales

The system shall allow authorized users to view sales transaction records.

7.6 Dashboard Requirements

FR-28 – Basic Dashboard

The system shall provide authorized users with basic summary information such as total medicines, available stock, low-stock items, and sales information.

8. NON-FUNCTIONAL REQUIREMENTS

8.1 Security

NFR-01 – Password Security

The system shall store passwords using secure password hashing rather than plain text.

NFR-02 – Authentication

Protected system functions shall require user authentication.

NFR-03 – Authorization

The system shall restrict functions according to user roles.

8.2 Performance

NFR-04 – Response Time

Under normal operating conditions, commonly used pages and requests should respond within approximately three seconds.

NFR-05 – Database Performance

The database shall be designed to support efficient retrieval and modification of pharmacy records.

8.3 Usability

NFR-06 – User Interface

The system shall provide a simple, understandable, and consistent user interface.

NFR-07 – Validation Messages

The system shall provide clear messages when input is invalid or an operation cannot be completed.

8.4 Reliability

NFR-08 – Data Integrity

The system shall maintain the integrity and consistency of stored pharmacy information.

NFR-09 – Error Handling

The system shall handle expected application errors without terminating unexpectedly.

8.5 Availability

NFR-10 – Availability

The system should be available during normal pharmacy operating hours when deployed in the target environment.

8.6 Maintainability

NFR-11 – Maintainable Architecture

The system shall use a modular structure that allows future changes with minimal unnecessary impact.

8.7 Compatibility

NFR-12 – Browser Compatibility

The web application should operate correctly on commonly used modern web browsers.

8.8 Responsiveness

NFR-13 – Responsive Interface

The user interface should adapt to common desktop, tablet, and mobile screen sizes.

9. USER STORIES

US-01 – Login

As a registered user, I want to log in using my credentials so that I can access the functions available to me.

US-02 – Manage Medicines

As an authorized user, I want to add and manage medicine records so that pharmacy information remains accurate.

US-03 – View Stock

As a pharmacist, I want to view stock information so that I can monitor medicine availability.

US-04 – Monitor Expiry

As a pharmacist, I want to identify expired or near-expiry medicines so that they can be handled appropriately.

US-05 – Manage Suppliers

As an authorized user, I want to manage supplier information so that supplier records are maintained.

US-06 – Record Sale

As a pharmacy staff member, I want to record a sale so that the transaction is stored and stock is updated.

US-07 – Search Medicine

As a pharmacy user, I want to search for medicines so that I can quickly find required information.

US-08 – View Dashboard

As a manager or authorized user, I want to view basic pharmacy summaries so that I can monitor operations.

US-09 – Logout

As an authenticated user, I want to log out so that my account is protected when I finish using the system.

10. ACCEPTANCE CRITERIA

AC-01 – Login

Given a registered user is on the login page, when valid credentials are entered and Login is selected, the system shall authenticate the user and display the appropriate system page. Invalid credentials shall produce an appropriate error message.

AC-02 – Add Medicine

Given an authorized user is logged in, when valid medicine information is entered and saved, the medicine record shall be stored and a success message shall be displayed.

AC-03 – Update Medicine

Given an authorized user is logged in, when valid medicine information is modified and updated, the database shall contain the updated information and a success message shall be displayed.

AC-04 – Stock Monitoring

Given medicine stock exists, when the stock quantity reaches the defined minimum level, the system shall identify the medicine as low stock.

AC-05 – Expiry Monitoring

Given medicines have expiry dates, when an expiry date has passed or meets the defined near-expiry condition, the system shall allow authorized users to identify the affected medicine.

AC-06 – Sale

Given a permitted user is authenticated, when a valid sale is completed, the sales record shall be saved, the total shall be calculated, and relevant stock quantities shall be reduced.

AC-07 – Search

Given an authenticated user enters a valid search value, the system shall display matching medicine records or an appropriate no-results message.

AC-08 – Logout

Given a user is authenticated, when Logout is selected, the authenticated session shall be terminated and the user shall be returned to the login page.

11. MOSCOW PRIORITIZATION

Priority	Requirements
Must Have	Login, Logout, Role-based access, Medicine CRUD, Stock Management, Sales, Supplier Management, Security
Should Have	Medicine Search, Low-stock Monitoring, Expiry Monitoring, Improved Validation, User Feedback
Could Have	Basic Dashboard, Basic sales summaries
Won't Have in v1.0	Online ordering, Payment Gateway, Mobile App, Advanced Accounting, External Integrations, Advanced Analytics

Note: "Won't have" means the functionality is excluded from Version 1.0 and may be considered for a future release.

12. PRODUCT BACKLOG

ID	Product Backlog Item	Priority	Related Requirements	Initial Status
US-01	User Login	Must	FR-01–FR-05	Planned
US-02	Manage Medicines	Must	FR-06–FR-12	Planned
US-03	View Stock	Must	FR-13, FR-18	Planned
US-04	Expiry Monitoring	Should	FR-15, FR-17	Planned
US-05	Manage Suppliers	Must	FR-19–FR-22	Planned
US-06	Record Sales	Must	FR-23–FR-27	Planned
US-07	Search Medicines	Should	FR-12	Planned
US-08	Basic Dashboard	Could	FR-28	Future / Planned
—	Online Ordering	Won't Have	—	Future Release
—	Payment Gateway	Won't Have	—	Future Release

13. Use Cases

Use Case ID	Use Case	Primary Actor
UC-01	Login	Administrator / Pharmacist / Pharmacy Staff
UC-02	Manage Medicines	Administrator / Pharmacist
UC-03	View Stock	Administrator / Pharmacist / Pharmacy Staff
UC-04	Monitor Expiry	Administrator / Pharmacist
UC-05	Manage Suppliers	Administrator / Pharmacist
UC-06	Record Sale	Pharmacist / Pharmacy Staff
UC-07	Search Medicine	Administrator / Pharmacist / Pharmacy Staff
UC-08	View Dashboard	Administrator / Pharmacist
UC-09	Logout	Administrator / Pharmacist / Pharmacy Staff

13.2 Use Case Example – Record Sale

Field	Description
Use Case ID	UC-06
Name	Record Sale
Actor	Pharmacist / Pharmacy Staff
Preconditions	User must be authenticated and have permission to record sales.
Trigger	User selects the sales function.
Main Flow	Search/select medicine → Enter quantity → Validate stock → Calculate total → Save sale → Deduct stock → Display confirmation
Alternative Flow	Invalid quantity, unavailable stock, or invalid medicine selection.
Postcondition	Sales transaction is stored and relevant stock is updated.

14. BUSINESS RULES

BR-01 – Only authenticated users may access protected system functionality.

BR-02 – Users may access only functions permitted by their assigned role.

BR-03 – Each medicine shall have a unique system identifier/code.

BR-04 – Required medicine information must be provided before a record can be saved.

BR-05 – Stock quantity shall not be reduced below the permitted available quantity.

BR-06 – A completed sale shall update the relevant stock quantity.

BR-07 – Medicine expiry dates shall be stored as valid dates.

BR-08 – Supplier information shall contain required contact details according to the system form.

BR-09 – Passwords must not be stored in plain text.

BR-10 – Sales totals shall be calculated from the medicines and quantities recorded in the transaction.

15. DATA REQUIREMENTS

15.1 User Information

- User ID
- Username
- Password Hash
- Role
- Account Status

15.2 Medicine Information

- Medicine ID / Code
- Medicine Name
- Category
- Description
- Unit Price
- Minimum Stock Level
- Status

15.3 Inventory / Batch Information

- Batch ID / Number
- Medicine ID
- Quantity
- Manufacturing Date (if required)
- Expiry Date
- Supplier ID

15.4 Supplier Information

- Supplier ID
- Supplier Name
- Contact Person
- Phone Number
- Email
- Address

15.5 Sales Information

- Sale ID
- Sale Date/Time
- User ID
- Total Amount
- Payment/Transaction Details as applicable

15.6 Sales Item Information

- Sale Item ID
- Sale ID
- Medicine ID
- Quantity
- Unit Price
- Line Total

Detailed database structures, relationships, normalization, constraints, and SQL schemas will be documented separately during the Database Design phase.

16. SYSTEM CONSTRAINTS

1. The application will be developed as a web-based system.
2. HTML, CSS, and JavaScript will be used for the frontend.
3. PHP will be used for the backend/server-side processing.
4. MySQL will be used as the database.
5. XAMPP will be used as the local development environment.
6. Visual Studio Code will be used as the code editor.
7. Version 1.0 will focus on essential pharmacy management functionality.
8. New requirements after baseline approval must follow the Change Request process.

17. ASSUMPTIONS

1. The pharmacy will provide authorized users to test the system.
2. Users will have access to a modern web browser.
3. Users will receive appropriate login credentials.
4. Medicine, supplier, and sales information entered into the system will be provided by authorized pharmacy personnel.
5. The final deployment environment will be defined at a later stage.
6. Future functionality can be introduced through controlled software changes.
7. Requirements may evolve after Version 1.0 and will therefore be managed through the Software Change Management process.

18. REQUIREMENTS TRACEABILITY MATRIX

Requirement	User Story	Acceptance Criteria	Test Case
FR-01 Login	US-01	AC-01	TC-01
FR-04 Logout	US-09	AC-08	TC-02
FR-06 Add Medicine	US-02	AC-02	TC-03
FR-10 Update Medicine	US-02	AC-03	TC-04
FR-13 Stock Management	US-03	AC-04	TC-05
FR-17 Expiry Monitoring	US-04	AC-05	TC-06
FR-19 Manage Supplier	US-05	—	TC-07
FR-23 Record Sale	US-06	AC-06	TC-08
FR-12 Search Medicine	US-07	AC-07	TC-09
FR-28 Dashboard	US-08	—	TC-10

The test case identifiers will be formally defined and expanded during the Test Planning and Testing sessions.

19. REQUIREMENTS REVIEW

Review Question

Can Pharmacy Staff manage all administrative functions?

Decision: No. Access shall be restricted according to role permissions.

Review Question

Is online ordering included in Version 1.0?

Decision: No. It is outside the Version 1.0 scope.

Review Question

Is payment gateway integration included?

Decision: No. It may be considered for a future release.

Review Question

Can unauthorized users access protected pharmacy information?

Decision: No. Authentication and authorization are required.

Review Question

Does a completed sale update stock?

Decision: Yes. Relevant stock quantities shall be reduced after a successful sale.

Review Question

Are advanced analytics included?

Decision: No. Advanced analytics are outside the Version 1.0 baseline.

20. VERSION 1.0 REQUIREMENTS BASELINE

After stakeholder review and approval, the requirements contained in this document shall become the Requirements Baseline v1.0. The baseline establishes the official reference point for Version 1.0 development.

- New requirements shall not be added directly to the Version 1.0 scope.
- Any new requirement must be submitted through the Software Change Management process.
- Approved changes shall be assessed for their impact on requirements, design, database, implementation, testing, and documentation.
- Relevant baseline and version information shall be updated when an approved change is implemented.

General change process: New Requirement → Change Request → Impact Analysis → Change Approval → Implementation → Testing → Code Review → Documentation Update → New Version / Release

21. CONFIGURATION AND VERSION CONTROL

The approved requirements document shall be maintained under version control.

Item	Value
Baseline Name	Requirements Baseline v1.0
Version	1.0
Status	Proposed for Review / Approved after Sign-off
Version Control	Git / GitHub (if used)
Baseline Date	
Approved By	

22. FUTURE CHANGE MANAGEMENT

Once the Version 1.0 requirements have been baselined, future requirements may be introduced through the formal change management process.

Example Future Request: "The pharmacy now requires online ordering functionality."

The request shall follow the formal change process: Customer Request → CR-001 → Impact Analysis → Project Approval → Requirements Update → Design Update → Database Update → Backend Update → Frontend Update → Testing → Release v1.1.

23. VERSION HISTORY

Version	Date	Description	Author	Status
0.1		Initial requirements draft		Draft
0.2		Requirements reviewed and revised		Review
1.0		Requirements baseline established		Approved

24. APPROVAL

By approving this document, the relevant stakeholders confirm that the requirements defined for Version 1.0 represent the agreed initial scope of the Pharmacy Management System.

Role	Name	Signature	Date
Client / Business Representative	_____	_____	_____
Project Team Representative	_____	_____	_____
Lecturer / Supervisor	_____	_____	_____

APPENDIX A – VERSION 1.0 SCOPE SUMMARY

Included

- Authentication
- Login / Logout
- Role-based access
- Medicine management
- Inventory / Stock management
- Batch and expiry management
- Supplier management
- Sales transactions
- Medicine search
- Low-stock monitoring
- Basic dashboard
- Basic security
- Responsive web interface
- PHP backend
- MySQL database

Not Included

- Online ordering
- Payment gateway
- Mobile application
- Advanced accounting
- Advanced analytics
- External integrations
- Automatic supplier purchasing
- SMS/email notification integration
- Multi-branch enterprise deployment
- Advanced audit logs

APPENDIX B – INITIAL TECHNOLOGY ENVIRONMENT

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Development Environment	XAMPP
Code Editor	Visual Studio Code
Version Control	Git / GitHub (if used)

APPENDIX C – PROJECT LIFECYCLE

Project Scenario → Requirements Engineering → Requirements Baseline → System Analysis & Design → Design Baseline → Database Design → Database Baseline → Backend Development → Frontend Development → Integration → Testing → Version 1.0 Release

After the first release: Change Request → Impact Analysis → Approved Change → Implementation → Testing → New Release

Appendix D – Baseline Statement

Requirements Baseline v1.0

The requirements contained in this document constitute the proposed Version 1.0 requirements baseline for the Pharmacy Management System. Any modification to these requirements after baseline approval must be controlled through the Software Change Management process.

Baseline Information	Value
Baseline Tag	requirements-v1.0
Baseline Date	
Approved By	
Approval Signature	

APPENDIX E – PROJECT DOCUMENT SEQUENCE

01 – Initial Client Scenario → 02 – Initial Project Proposal → 03 – Project Plan / Practical Roadmap → 04 – SRS + Requirements Baseline v1.0 → SCM Setup → System Design → Database Design → Development → Testing → Version 1.0 Release → CR-001 → Impact Analysis → Version 1.1

FINAL BASELINE NOTE

The Version 1.0 scope has intentionally been kept focused on authentication, medicine and inventory management, supplier management, sales processing, and role-based access. Features such as online ordering, payment gateway integration, mobile application support, advanced accounting, and advanced analytics are excluded from the Version 1.0 baseline. These can later be introduced through formal Change Requests.